

Production Supporting Systems in Factories

ระบบสนับสนุนการผลิตในโรงงานอุตสาหกรรม

Context

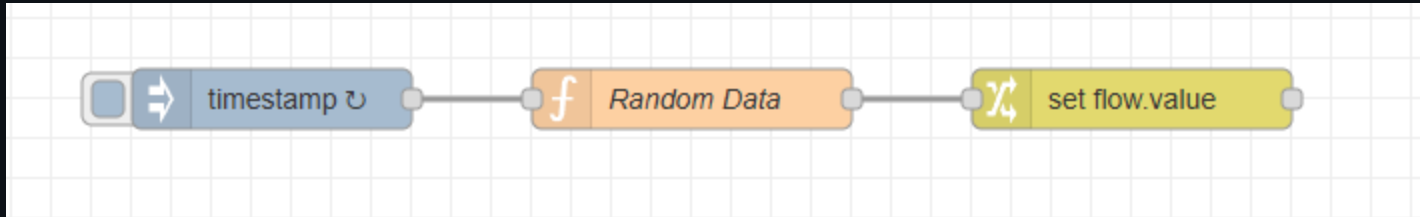
Scope

The `scope` of a particular context value.

- `Node` - only visible to the node that set the value
- `Flow` - visible to all nodes on the same flow (or tab in the editor)
- `Global` - visible to all nodes

Use `flow` Context to Pass Values

Set flow Context



inject Node

Edit inject node

Delete Cancel Done

Properties

Name

msg. payload = milliseconds since epoch

msg. topic = a_z

+ add inject now

☐ Inject once after 0.1 seconds, then

Repeat interval

every 1 seconds

random data Function Node

```
msg.payload = Math.random();  
return msg;
```

change Node

Edit change node

Delete Cancel Done

⚙ Properties

📌 Name Name

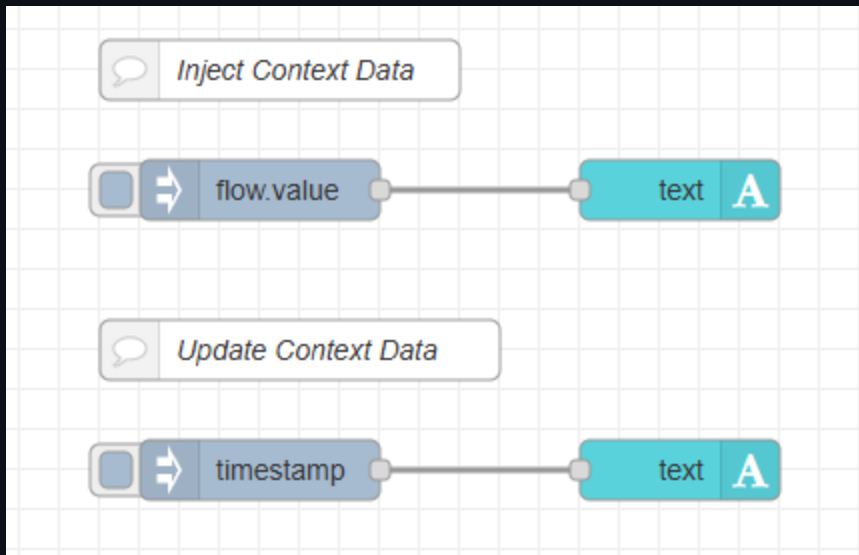
☰ Rules

Set ▼ ▼ flow. value

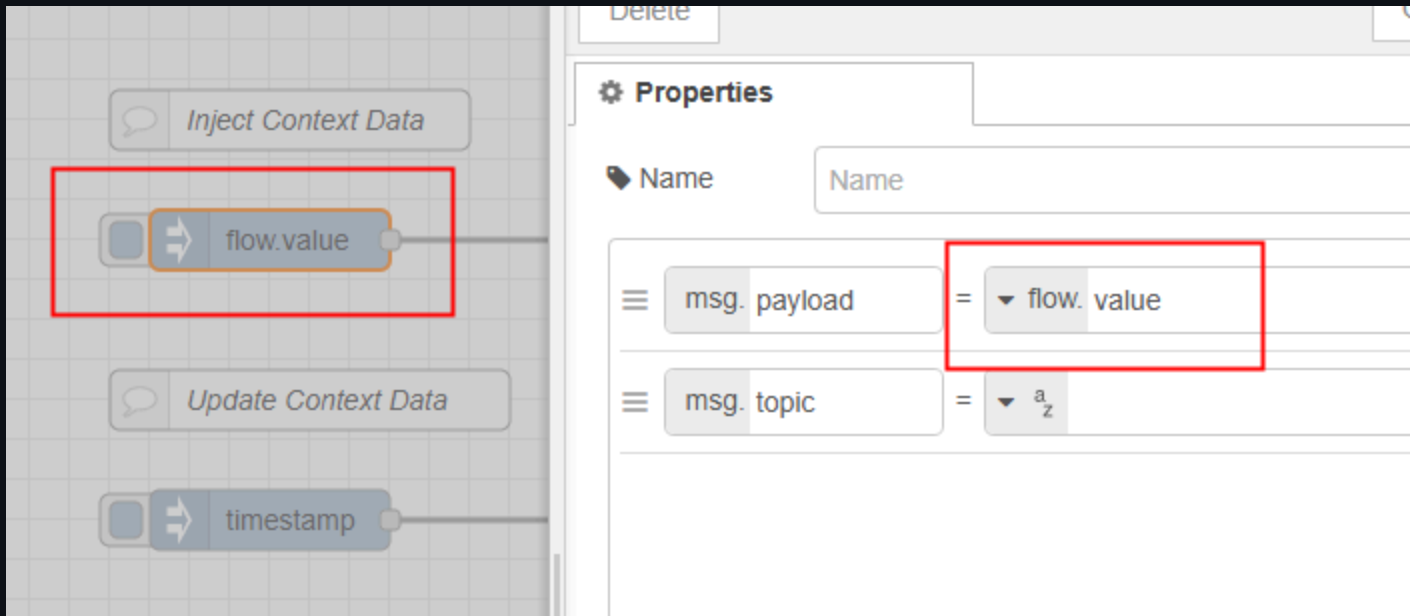
☰ to the value ▼ msg. payload ✕

☐ Deep copy value

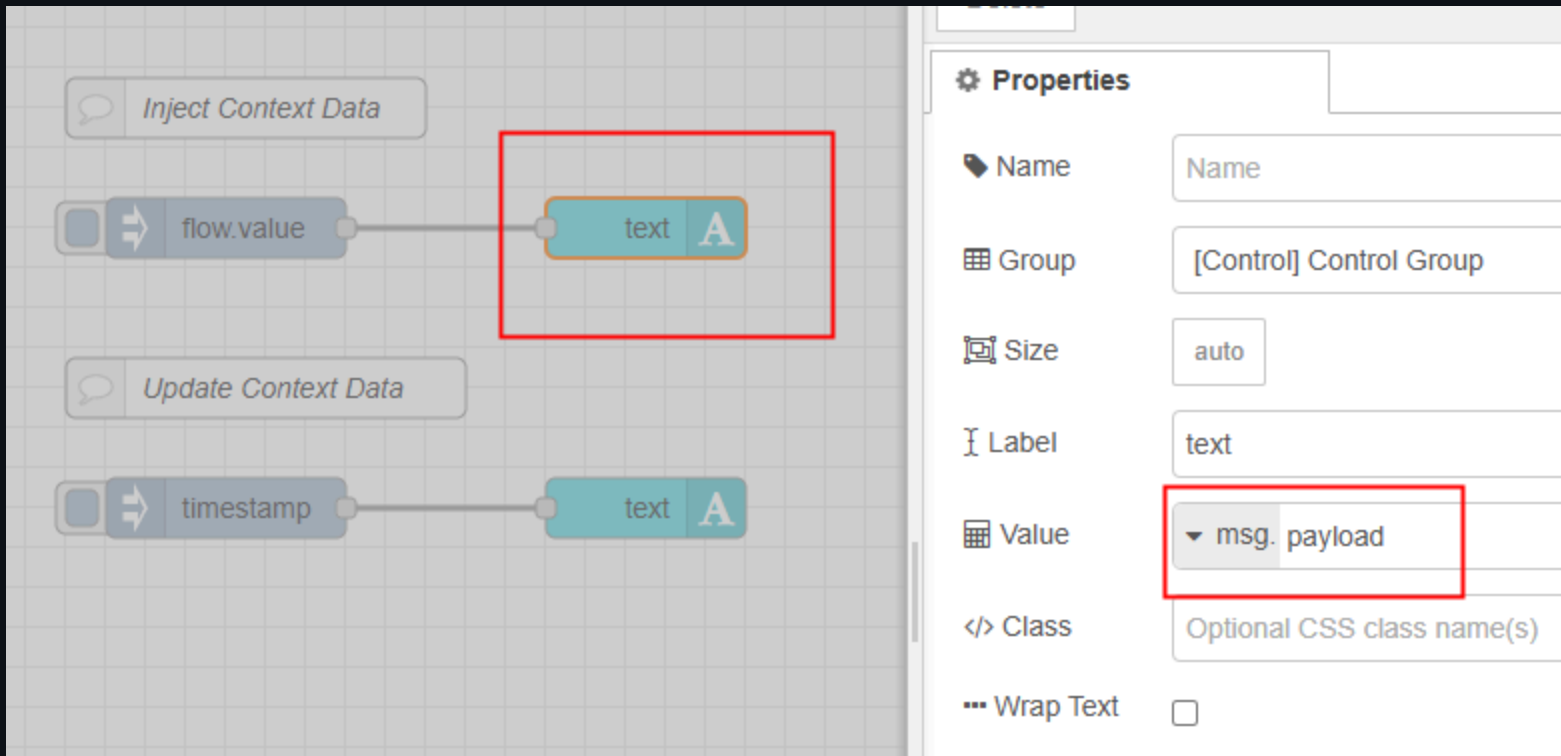
Get Context Data



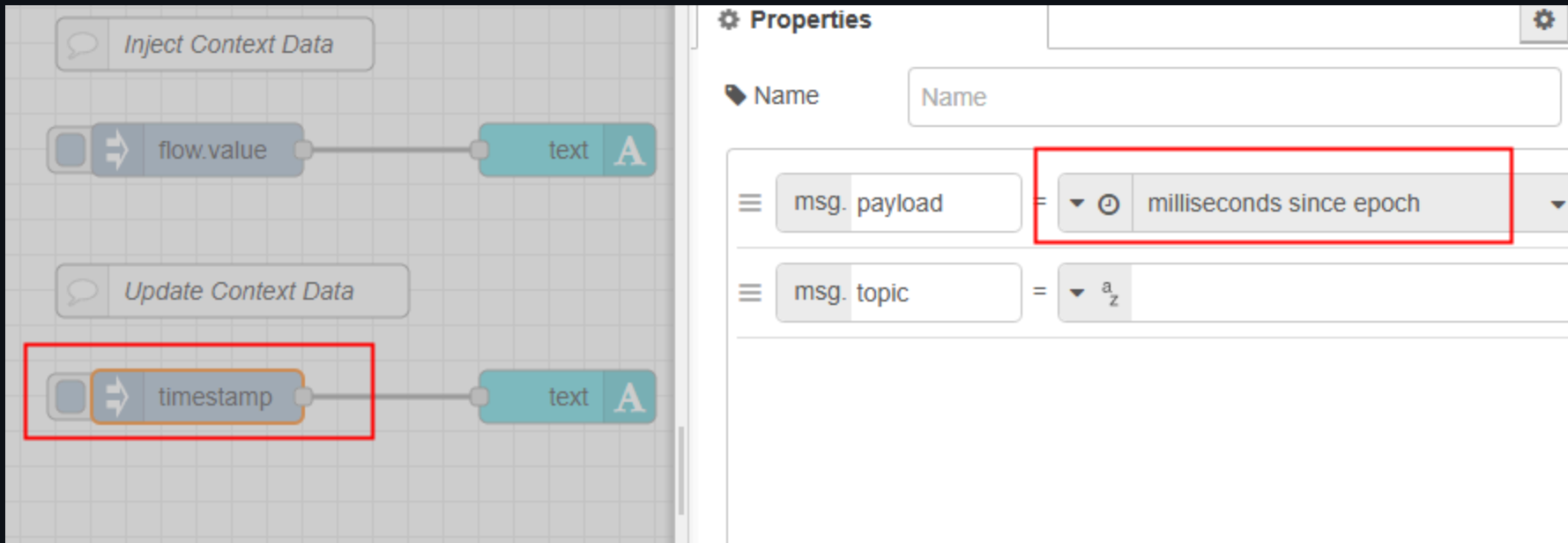
Inject Context Data Flow



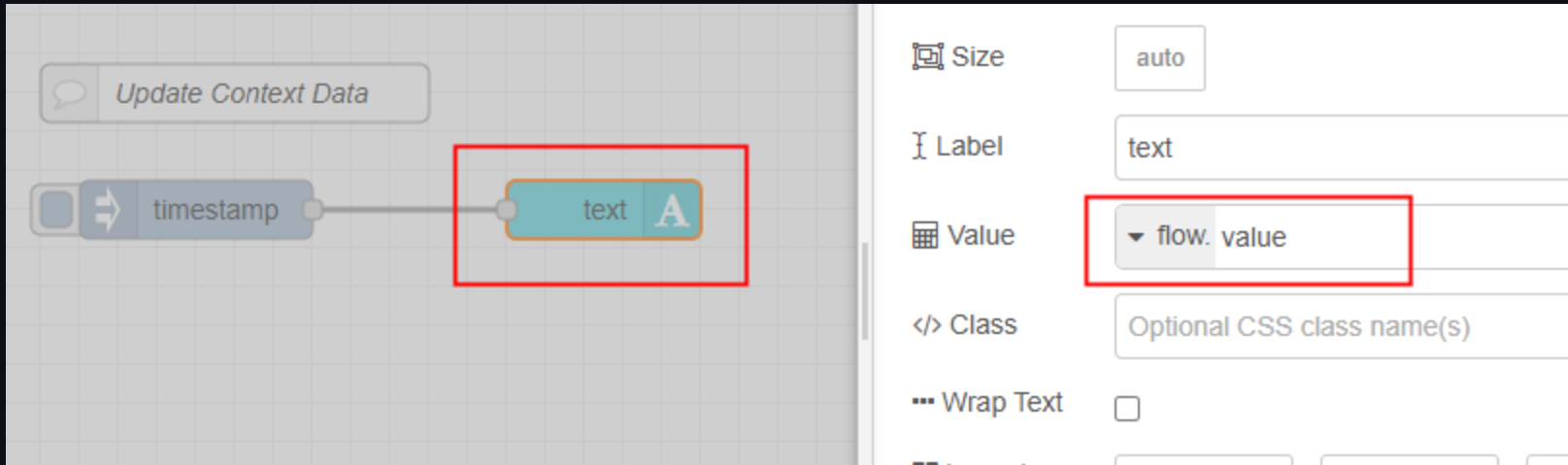
Inject Context Data Flow



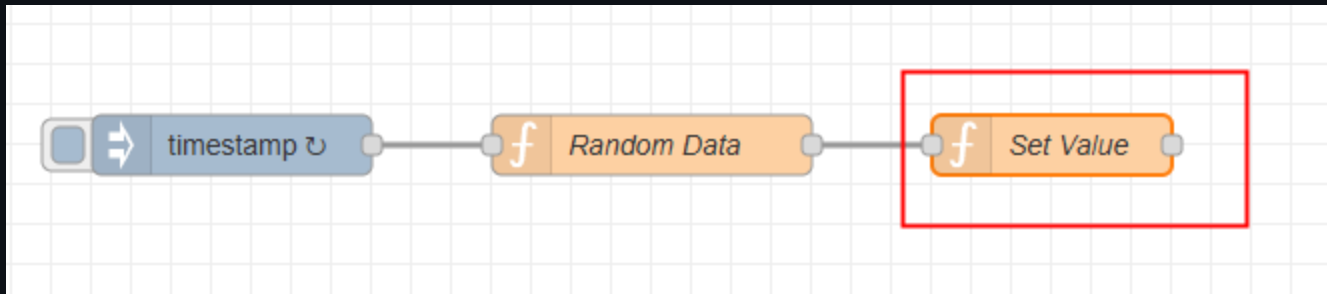
Update Context Data Flow



Update Context Data Flow

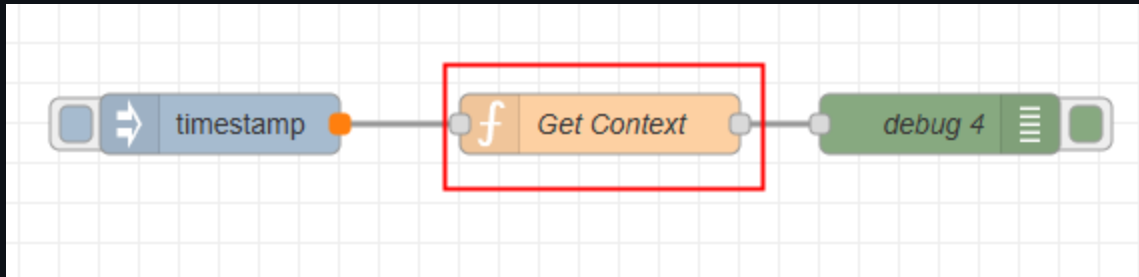


Using **function** Node to Set Context Value



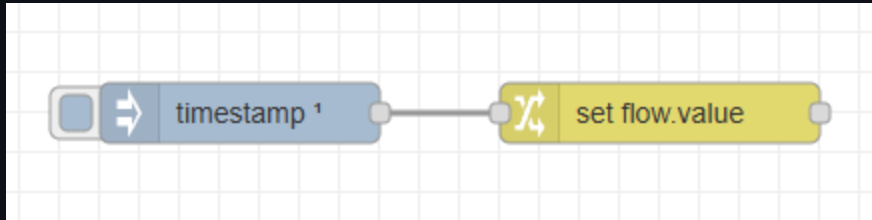
```
flow.set("value", msg.payload);
```

Using **function** Node to Get Context Value

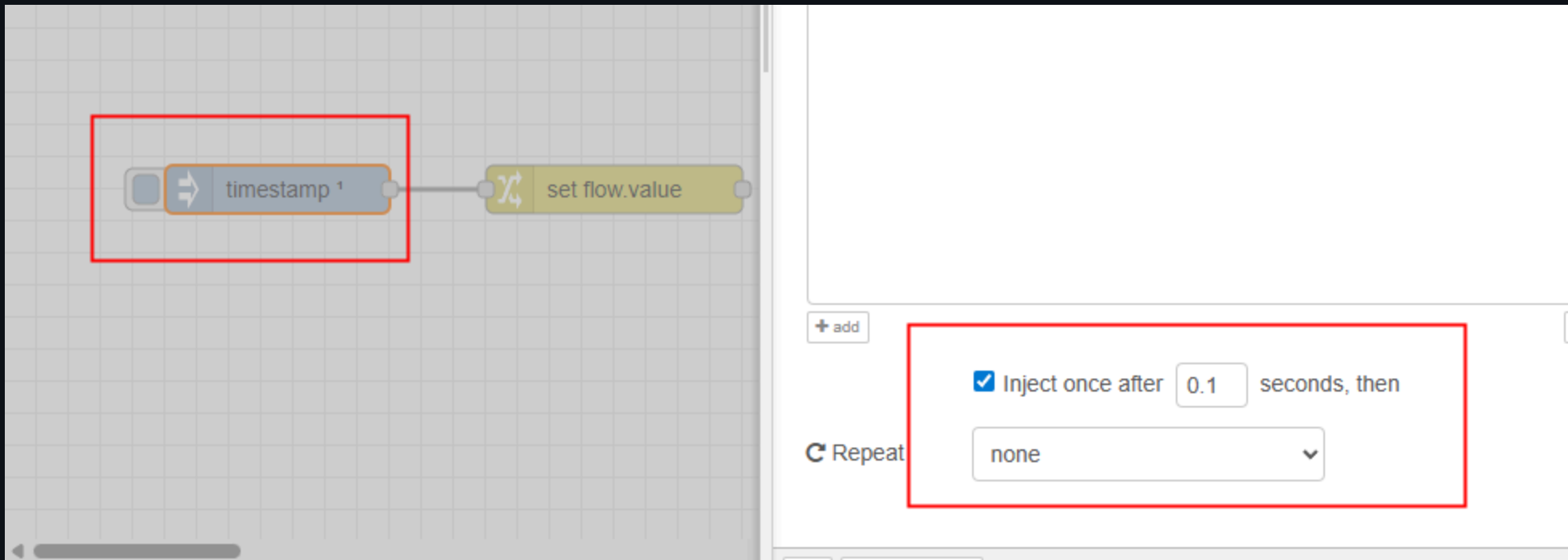


```
const value = flow.get("value") ?? 0;  
msg.payload = value;  
return msg;
```

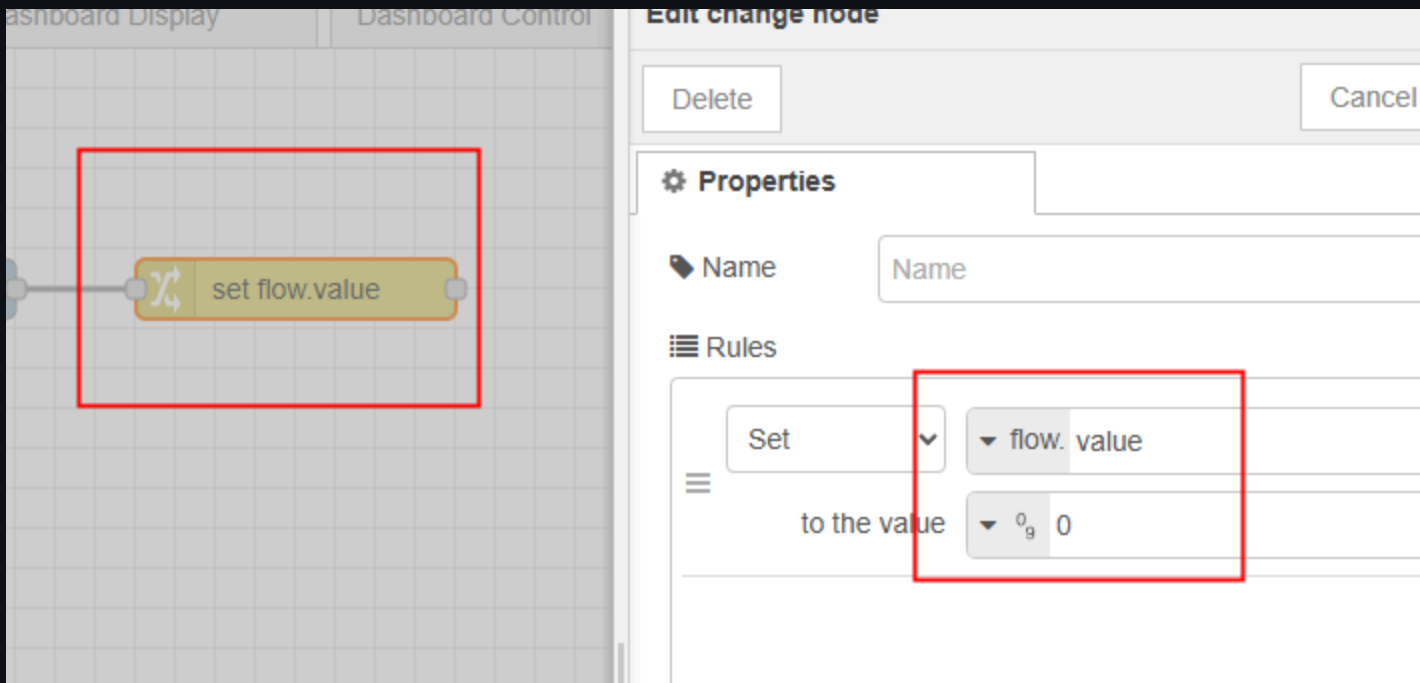
Initialize Context Value



inject Node

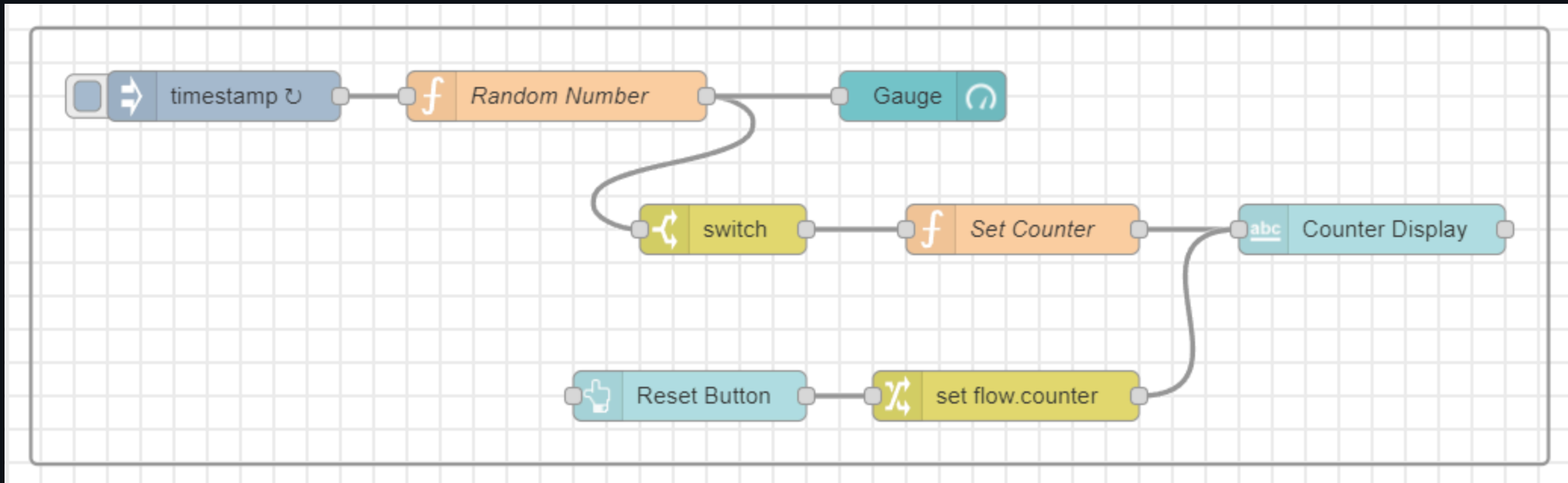


change Node



Counter with reset button

Flow



Random Number (function) node

```
msg.payload = Math.random();  
return msg;
```

Switch node

Edit switch node

Delete

Cancel

Done

⚙ Properties

⚙

📄

🔍

📌 Name

Name

⋮ Property

▼ msg. payload

☰

>=

▼

▼ 0.5

0.5

→ 1

×

+ add

checking all rules

▼

☐ recreate message sequences

📄

○ Enabled

Set Counter (Function) node

```
const curCounter = flow.get("counter") ?? 0;  
flow.set("counter", curCounter + 1);  
msg.payload = curCounter + 1;  
return msg;
```

button node

Edit button node

Delete Cancel Done

Properties

Group [Context] Group 1

Size auto

Icon optional icon

Label Reset Button

Tooltip optional tooltip

Color optional text/icon color

Background optional background color

☒ When clicked, send:

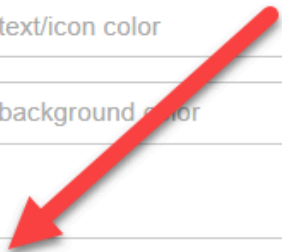
Payload 0

Topic msg.topic

☐ If msg arrives on input, emulate a button click:

Class Optional CSS class name(s) for widget

Name Name



change node

Edit change node

Delete Cancel Done

⚙ Properties

🏷 Name Name

☰ Rules

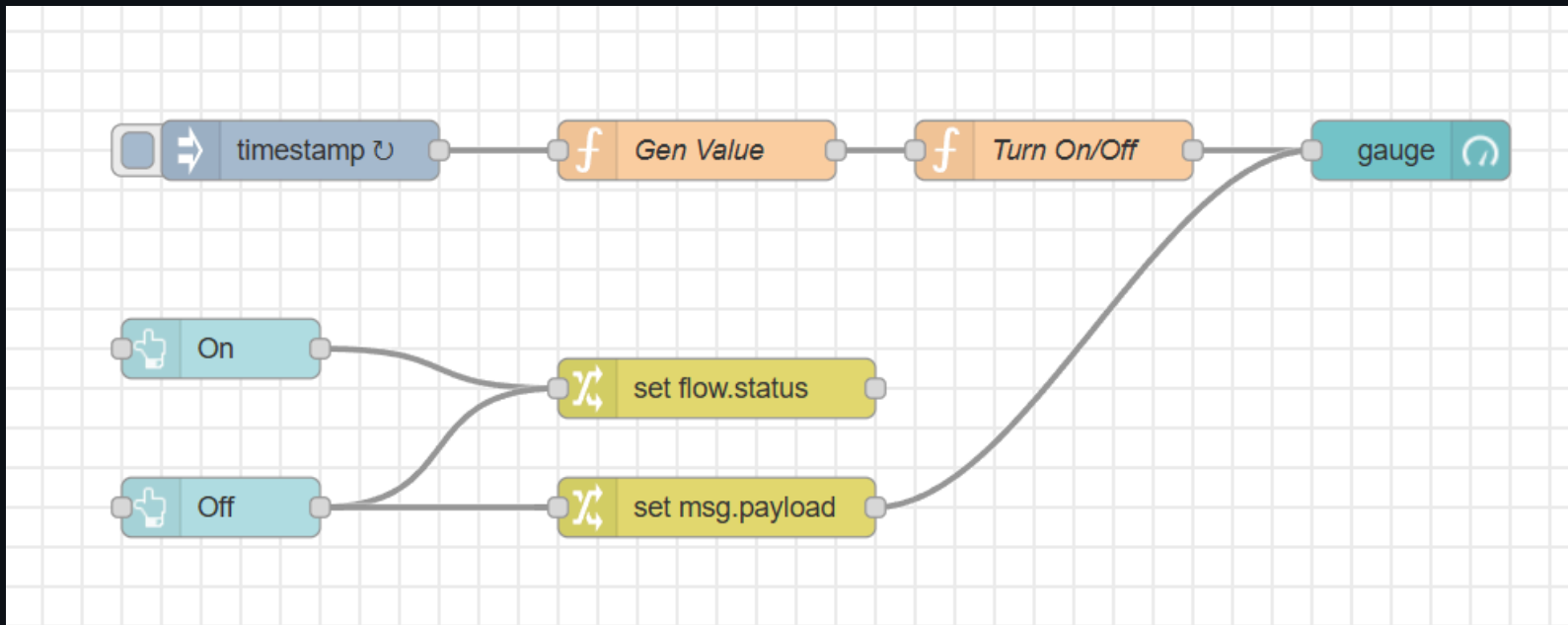
Set ▼ ▼ flow. counter

☰ to the value ▼ msg. payload ✕

☐ Deep copy value

Turning sensor display on and off

Flow

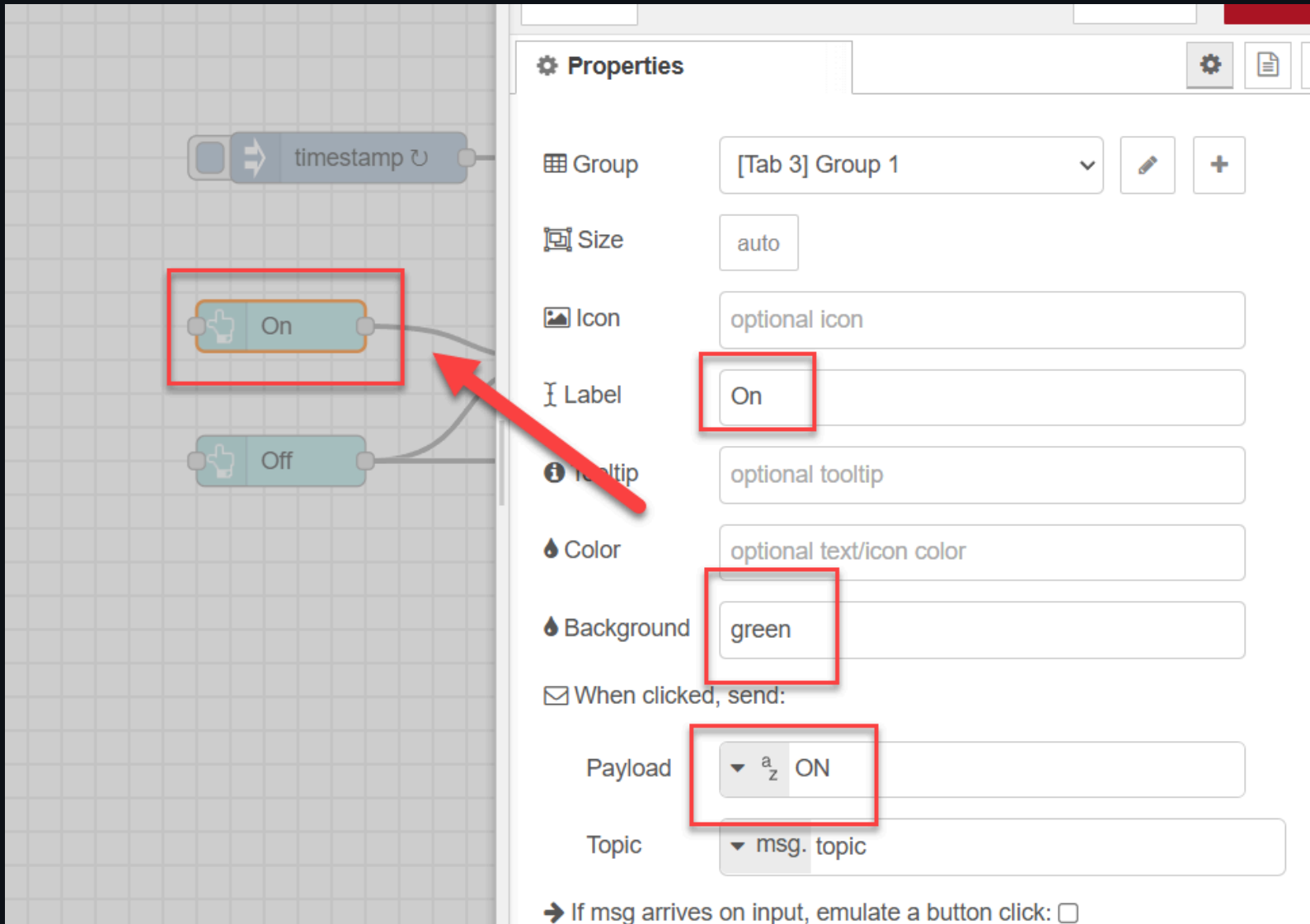


function node (Gen Value)

```
msg.payload = Math.random();  
return msg;
```

function node (Turn On/Off)

```
const status = flow.get("status") ?? "OFF";  
if (status === "ON") {  
    return msg;  
} else {  
    // Do nothing  
}
```



timestamp

On

Off

Properties

Group [Tab 3] Group 1

Size auto

Icon optional icon

Label **On**

Tooltip optional tooltip

Color optional text/icon color

Background **green**

When clicked, send:

Payload **a_z ON**

Topic msg. topic

→ If msg arrives on input, emulate a button click: ☐

timestamp ↻

On

Off

Properties

Group [Tab 3] Group 1

Size auto

Icon optional icon

Label Off

Tooltip optional tooltip

Color optional text/icon color

Background red

When clicked, send:

Payload a_z OFF

Topic msg. topic

→ If msg arrives on input, emulate a button click: ☐

The image shows a Node-RED interface. On the left, a canvas displays three nodes: 'Gen Value' (orange), 'set flow.status' (yellow), and 'set msg.payload' (yellow). The 'set flow.status' node is highlighted with a red rectangle. A red arrow points from this node to the 'Properties' sidebar on the right. The sidebar has a 'Name' field and a 'Rules' section. In the 'Rules' section, a dropdown menu is set to 'Set', and the text 'to the value' is visible. A red rectangle highlights the 'flow.status' and 'msg.payload' options under the dropdown, along with an unchecked 'Deep copy value' checkbox.

Properties

Name Name

Rules

Set ▼

to the value

▼ flow. status

▼ msg. payload

☐ Deep copy value

